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(54) **METHOD FOR SWITCHING AN ELECTRICALLY SHIFTABLE TRANSMISSION FOR A VEHICLE, AND ELECTRICALLY SHIFTABLE TRANSMISSION**

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***F16H 61/02*** (2006.01)

(52) **U.S. Cl.**
CPC ................................ ***F16H 61/0213*** (2013.01)

(58) **Field of Classification Search**
CPC .... F16H 61/0213; F16D 48/064; F16D 48/10; F16D 2500/10462
See application file for complete search history.

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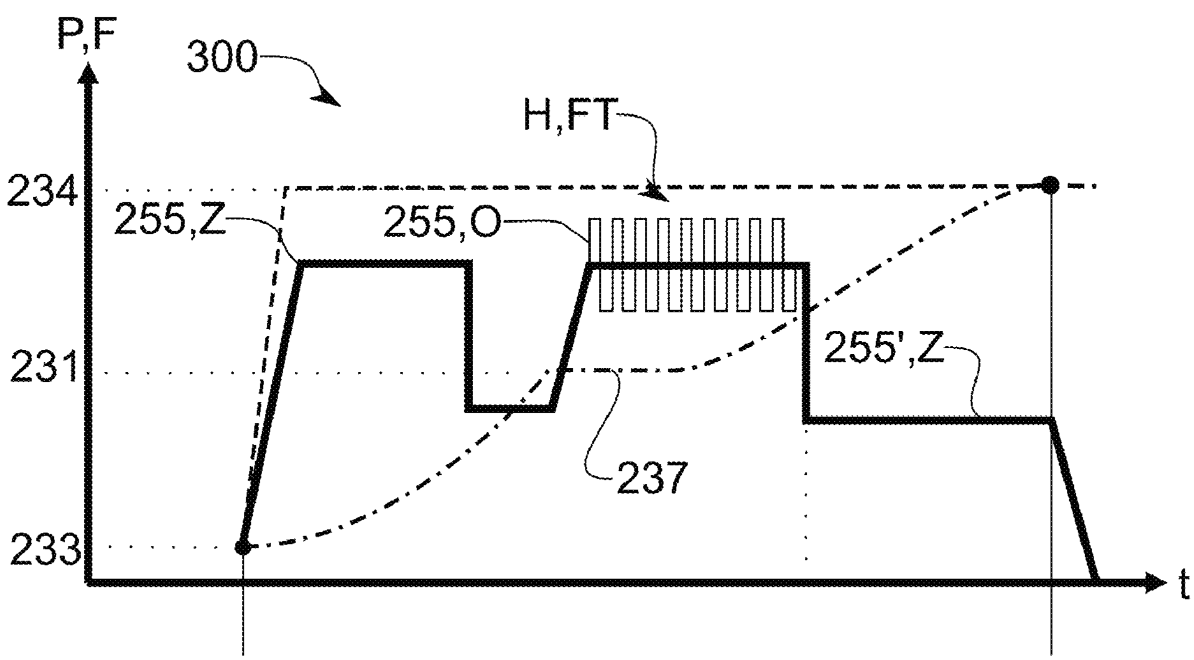
(57) **ABSTRACT**

A method (**100**) for shifting an electrically shiftable transmission (**230**) for a vehicle (**200***a*), in particular a commercial vehicle (**200***b*) includes detecting (**110**) a state (**237**) of a shifting element (**235**) relating to a blocking position (**231**) of a shifting element (**235**) relative to a further shifting element (**236**). A control signal (**255**) for controlling a movement of the shifting element (**235**) is determined, taking into account the detection of the blocking position (**231**). The control signal (**255**) defines a target variable (Z) of the movement of the shifting element (**235**) and an oscillation (O) about the target variable (Z). The control signal (**255**) is output and moves the shifting element (**235**) out of the blocking position and into a final position of engagement with the further shifting element (**236**).

**20 Claims, 2 Drawing Sheets**



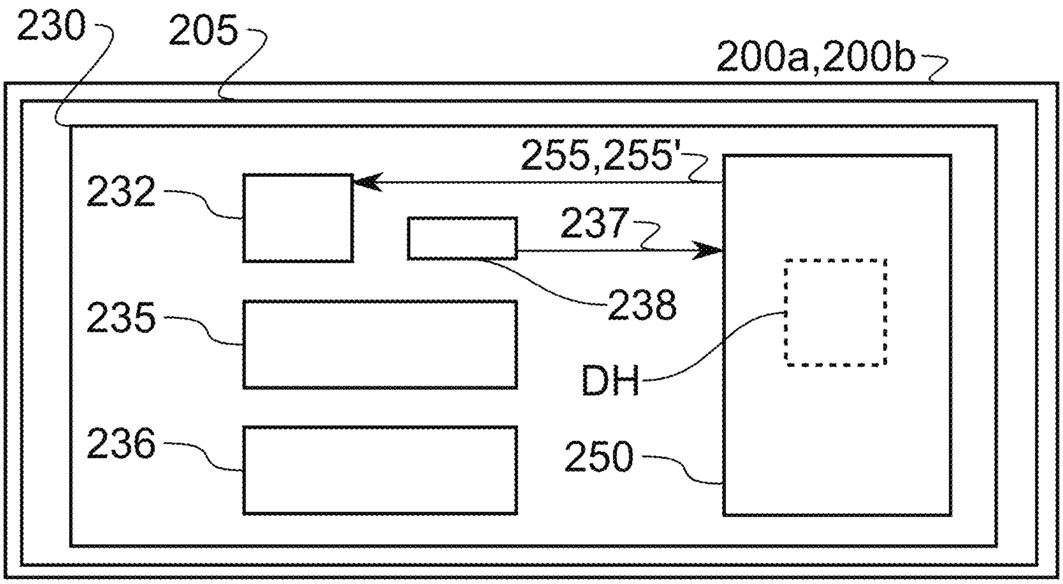

230
205
200a,200b
255,255'
232
237
238
235
DH
236
250

Fig.1

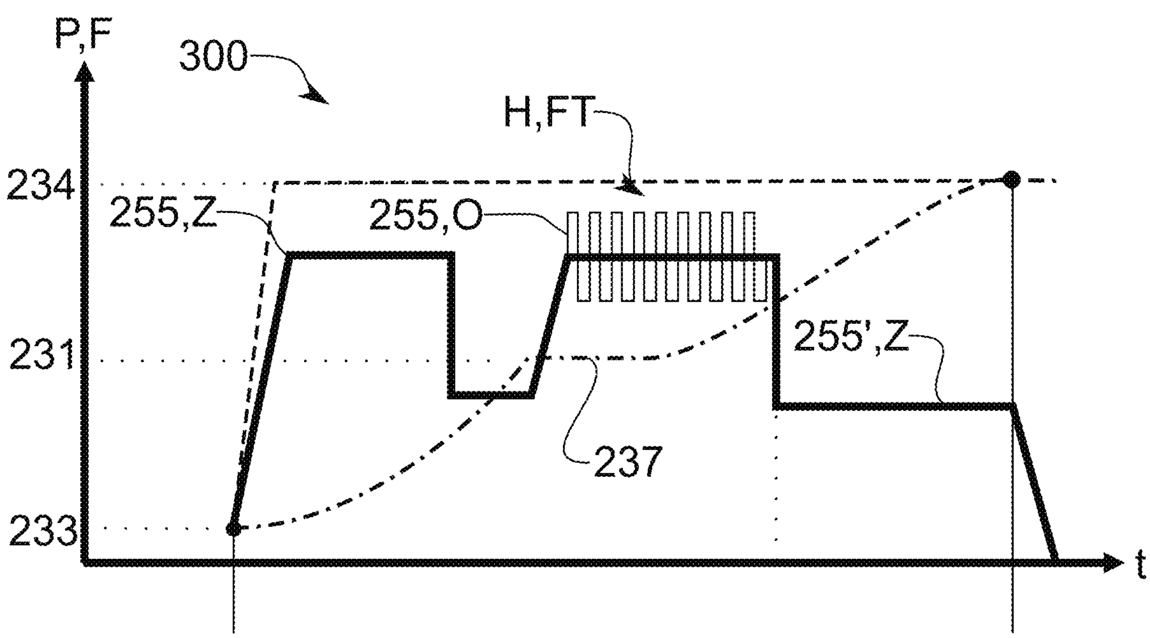

P,F
300
H,FT
234
255,Z
255,O
231
255',Z
237
233
t

Fig.2

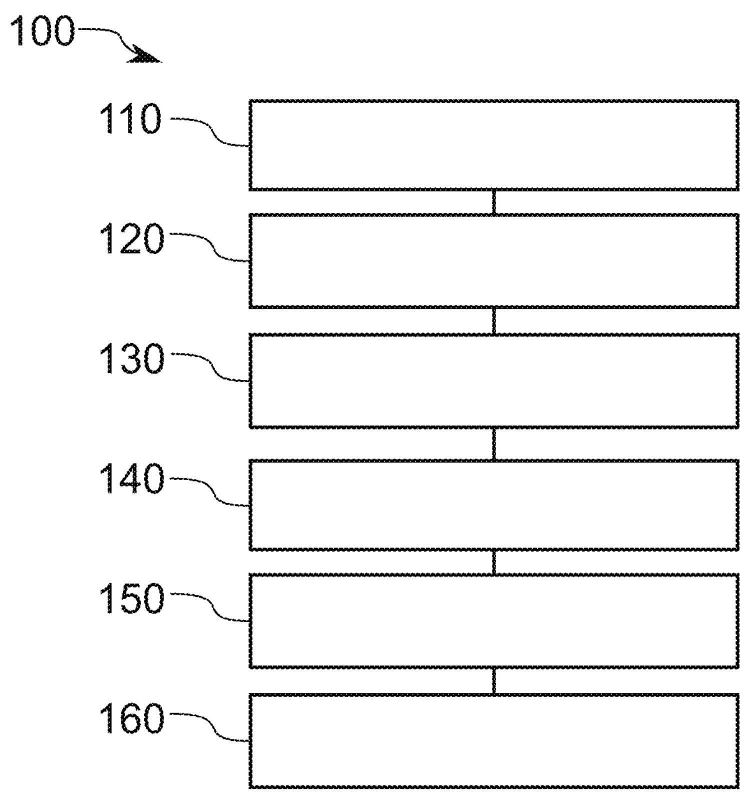



Fig.3

# METHOD FOR SWITCHING AN ELECTRICALLY SHIFTABLE TRANSMISSION FOR A VEHICLE, AND ELECTRICALLY SHIFTABLE TRANSMISSION

## FIELD

The present disclosure relates to a method for shifting an electrically shiftable transmission for a vehicle, in particular a commercial vehicle. The present disclosure also relates to a computer program and/or computer-readable medium, a control device for a vehicle, in particular a commercial vehicle, with an electrically shiftable transmission, an electrically shiftable transmission with a shifting element, an axle arrangement for a vehicle, in particular a commercial vehicle, and a vehicle, in particular a commercial vehicle.

## BACKGROUND

So-called dog clutches for shifting transmissions are particularly well known for commercial vehicles. To accomplish a gear shift, a shifting element of the transmission is moved by an actuator to be brought into operative connection with another shifting element and/or to resolve such an operative connection. In particular, in pneumatically shiftable transmissions, blocking positions are known to be a typical and recurring condition in which the two shifting elements are, for example, in a tooth-to-tooth position, which hinders meshing movement. Furthermore, particularly under load, decoupling of two intermeshing shifting elements can be made difficult and/or practically impossible by torques acting between the shifting elements (so-called "sticking") and thus a blocking position can occur.

To resolve a blocking position, typically in a pneumatically shiftable transmission, a force for moving the shifting element is increased to resolve the blocking position and, if the blocking position cannot be eliminated, the force is removed and the shifting process is attempted again.

A blocking position can lead to a delay in the shifting time, because increasing the force, eliminating the blocking position, and a possible repeat of the shifting process take time. Furthermore, such blocking positions can lead to a reduction in the service life of the transmission elements. The forces generated during gear shifting can result in high mechanical stresses, which means that the transmission typically has to be designed in a complex and elaborate way to withstand the forces.

Electrically shiftable transmissions are known from the prior art. A target position and/or a movement of an electric shift actuator is set similar to the target position of a pneumatic shift actuator. The target position is specified and the shift actuator adjusts a movement of the shifting element of the transmission according to the demand using a force.

DE 694 02 779 T2 discloses a method for controlling a force exerted on a shifting mechanism of an automatic mechanical transmission during a shifting process, the transmission having at least one shift rail, the shifting mechanism including a shifting finger driven by a motor, the shifting finger cooperating with the shift rail to effect shifting of the transmission, the method comprising energizing a motor with a pulse width modulated control signal with variable duty cycle to generate a target current which is applied to the motor and detecting the current drawn by the motor, continuously readjusting the duty cycle of the control signal during a shifting process as a function of the error sum between the detected current and the target current plus the



rate of change of the error to reduce the error to zero. The motor is connected to a direct current source, such as, e.g., the vehicle battery.

Control electronics carry out all relevant computing operations and ensure communication with other electronic systems in the vehicle. Said control electronics control the motor according to the described method and convert the direct current, e.g., from the vehicle battery, into suitable phase currents of the motor.

## SUMMARY

The underlying object of the present disclosure is to provide an improved method for shifting an electrically shiftable transmission. In particular, the present disclosure can achieve the object of effectively eliminating and/or preventing a blocking position.

The object is achieved by the method and the subject matter according to the present disclosure. Further developments of the present disclosure are described herein.

According to the present disclosure, a method for shifting an electrically shiftable transmission for a vehicle, in particular a commercial vehicle, is provided. The method includes: detecting a state of a shifting element of the transmission relating to a blocking position of the shifting element; determining a control signal for controlling a movement of the shifting element taking into account the detected state and/or the blocking position, wherein the control signal defines a target variable of the movement of the shifting element and an oscillation about the target variable; and outputting the control signal and moving the shifting element based on the control signal.

It was recognized that with an electrically shiftable transmission it is possible to detect and control the movement of the shifting element in a more differentiated manner than with, for example, a pneumatically shiftable transmission. This makes it possible to detect the state of the shifting element in which the blocking position is detected and/or can be predicted. For example, the state of the shifting element can characterize a movement of the shifting element caused by a shift actuator for moving the shifting element.

To resolve and/or prevent the blocking position, the movement of the shifting element is adjusted by determining the control signal, such that the oscillations or pulses are imposed on a target variable characterizing the movement. In other words, the movement of the shifting element is controlled according to the target variable. In addition to the target variable, the oscillation is imposed on the control signal. The oscillations may be any temporal change of the target variable, which, in particular alternatingly, increases and decreases the target variable. In other words, the oscillation may be a harmonic oscillation, may be composed of a plurality of harmonic oscillations, and/or may be a non-harmonic oscillation.

The present disclosure has recognized that it is thus possible to generate a dynamic force between the shifting element and the further shifting element, which leads to a higher probability of eliminating the blocking position. The oscillation can cause micro-movements that resolve and/or prevent the blocking position. This improves the shifting performance and reduces the time required to perform a shifting process. High mechanical stress on the transmission caused by excessive force can thus be prevented. This enables a more targeted and effective design of the transmission components, which can lead to cost reduction.

Optionally, the state of the shifting element indicates the blocking position. This makes it possible to effectively characterize the blocking position. For example, the state of the shifting element may be defined by a position of the shifting element which can be detected via a position sensor and optionally does not change over a predetermined time interval. Alternatively, or additionally, the state of the shifting element indicates the blocking position resulting from a movement of the shifting element. It was recognized that a blocking position can be prevented if the blocking position is predicted based on the movement of the shifting element. The movement of the shifting element can be detected to determine, based on a prediction of the position of the shifting element, whether a further movement of the shifting element could potentially result in a blocking position.

Optionally, the target variable includes a force, a velocity, and/or a position. This allows the target variable to be controlled depending on the application. The velocity and/or position can be effectively controlled by the electrical shiftability of the transmission.

Optionally, the oscillation defines a frequency of the movement of the shifting element. The oscillation may be a harmonic or sinusoidal oscillation. The frequency of the movement may be a characteristic, a minimum, and/or a maximum frequency. Optionally, the oscillation may define a plurality of frequencies of the movement of the shifting element to achieve more complex oscillations with different time scales to achieve different micro-movements.

Optionally, the oscillation is dependent on a speed difference between the shifting element and a further shifting element that can be brought into operative connection with the shifting element. This means that the oscillation of the movement of the shifting element may depend on the speed difference between the two shifting elements. This allows two variables characterizing the kinematics of the shifting process to be coordinated with one another. In particular, a frequency and/or an amplitude of the oscillation may be dependent on the speed difference.

Optionally, the determination of the control signal is carried out such that a force on the shifting element falls below a force threshold. This makes it possible to limit the oscillation. By falling below the force threshold, it is possible to ensure that no excessive forces act on components of the transmission, which can have a positive influence on the design and layout of the transmission. Accordingly, the force threshold may be selected by the design of the transmission and/or the design of the transmission may be adjusted to a force threshold to be achieved.

Optionally, the method includes: detecting the vacating of the blocking position; determining an adjusted control signal for controlling the movement of the shifting element after the blocking position is vacated, wherein the adjusted control signal defines a target variable of the movement of the shifting element without the oscillation; and, outputting the adjusted control signal for moving the shifting element based on the adjusted control signal. It was recognized that it is possible to transition to oscillation-free control of the shifting element after the blocking position has been resolved to enable effective shifting.

According to a further aspect of the present disclosure, a computer program and/or computer-readable medium is provided. The computer program and/or computer-readable medium comprises commands which, when the program or commands are executed by a computer, cause the computer to carry out the method and/or the steps of the method described herein. The computer program and/or computer-readable medium may include commands to carry out steps of the method described as optional to achieve a corresponding technical effect.

According to a further aspect of the present disclosure, a control device for a vehicle, in particular a commercial vehicle, with an electrically shiftable transmission is provided. The control device may be configured to carry out steps of the method described as optional to achieve a corresponding technical effect.

According to a further aspect of the present disclosure, an electrically shiftable transmission is provided. The electrically shiftable transmission has a shifting element and the control device described above. Optionally, the electrically shiftable transmission is an automated manual transmission (AMT).

According to one aspect of the present disclosure, an axle arrangement for a vehicle, in particular a commercial vehicle, is provided. The axle arrangement has the electrically shiftable transmission described above. The axle arrangement and/or the control device of the electrically shiftable transmission may be configured to carry out steps of the method described as optional and/or advantageous to achieve a corresponding technical effect.

According to a further aspect of the present disclosure, a vehicle, in particular a commercial vehicle, is provided. The vehicle has an electrically shiftable transmission with the control device described herein and/or the axle arrangement described above. The vehicle and/or the control device may be configured to carry out steps of the method described as optional and/or advantageous to achieve a corresponding technical effect.

## BRIEF DESCRIPTION OF THE DRAWINGS

Further advantages and features of the present disclosure as well as their technical effects can be derived from the figures and the description of the preferred embodiments shown in the figures. In the figures:

FIG. **1** is a schematic illustration of a vehicle, in particular a commercial vehicle, according to an embodiment of the present disclosure;

FIG. **2** is a schematic illustration of a movement profile for moving a shifting element of an electrically shiftable transmission according to an embodiment of the present disclosure; and

FIG. **3** is a schematic illustration of a sequence of a method according to an embodiment of the present disclosure.

## DETAILED DESCRIPTION

FIG. **1** is a schematic illustration of a vehicle **200***a*, in particular a commercial vehicle **200***b*, according to an embodiment of the present disclosure. The vehicle **200***a*, in particular a commercial vehicle **200***b*, is hereinafter referred to as vehicle **200***a*, **200***b*. The vehicle **200***a*, **200***b* is a land vehicle. The vehicle **200***a*, **200***b* is, for example, a truck, a bus and/or a tractor unit of a multi-unit vehicle.

The vehicle **200***a*, **200***b* is configured to carry out the method **100** described with reference to FIG. **3**. For this purpose, the vehicle **200***a*, **200***b* comprises an electrically shiftable transmission **230**. The electrically shiftable transmission **230** has a shifting element **235** and a further shifting element **236**. A shifting process is achieved, for example, by bringing the shifting element **235** from a neutral position **233** into engagement with the further shifting element **236** in an end position **234** (see FIG. **2**). In this case, a distance between the shifting element **235** and the further shifting element **236** changes. A variable describing the distance between the shifting element **235** and the further shifting element **236** is referred to below as position P. Each of the shifting elements **235**, **236** can rotate at a speed. A speed difference DH can occur between the shifting element **235** and the second shifting element **236**.

To set and/or control the position P, the transmission **230** comprises a control device **250** and a controllable electric shift actuator **232** for actuating the shifting element **235**. The controllable electric shift actuator **232** comprises, for example, an electronically commutated synchronous motor or a brushless DC motor as a servomotor.

The control device **250** is further configured to carry out the method **100** described with reference to FIG. **3** to carry out a shifting of the electrically shiftable transmission **230**.

For this purpose, the transmission **230** has a position sensor **238** connected to the control device **250**. The position sensor **238** is configured to detect the position P of the shifting element **235**. The control device **250** is configured to receive the position P from the position sensor **238**. Based on the position P, the control device **250** can detect the movement of the shifting element **235** and a state **237** relating to a blocking position **231** (see FIG. **2**) of the shifting element **235**. The state **237** can indicate the position P of the shifting element **235** in the blocking position **231** and/or a blocking position **231** resulting from a movement of the shifting element **235**. Furthermore, the detected position P can be used to determine whether the blocking position **231** has been vacated. The control device **250** is configured to detect and/or determine the speed difference DH between the shifting element **235** and the second shifting element **236**.

The control device **250** is configured to determine a control signal **255** and an adjusted control signal **255'**. The control signal **255** for controlling the movement of the shifting element **235** is determined taking into account the state **237** and/or the blocking position **231**, wherein the control signal **255** defines a target variable Z of the movement of the shifting element **235** and an oscillation O about the target variable Z (see FIG. **2**). When it has been detected that the blocking position **231** has been vacated, the adjusted control signal **255'** can be determined for controlling the movement of the shifting element **235** after the blocking position **231** has been vacated, wherein the adjusted control signal **255'** defines a target variable Z of the movement of the shifting element **235** without the oscillation O (see FIG. **2**). The control signal **255** and the adjusted control signal **255'** are described with reference to the movement profile **300** according to FIG. **2**.

As shown in FIG. **1**, the control device **250** and the shift actuator **232** are connected to one another by way of communication technology so that the control device **250** can transmit or output the control signal **255** and the adjusted control signal **255'** to the shift actuator **232** for moving the shifting element **235**.

FIG. **2** shows a schematic illustration of a movement profile **300** for moving a shifting element **235** of an electrically shiftable transmission **230** according to an embodiment of the present disclosure. Such a transmission **230** is described with reference to FIG. **1**. FIG. **2** is described with reference to FIG. **1** and its description.

The movement profile **300** according to FIG. **2** defines a relationship between the position P and a force F as a function of time t, in each case on an arbitrary scale. Time t is plotted as ordinate (value on the x-axis) and the position P and the force F are plotted as abscissa (value on the y-axis). A neutral position **233**, a blocking position **231**, and an end position **234** are illustrated as position P.

In a case without blocking in the blocking position **231**, the shifting element **235** can be moved from the neutral position **233** to the end position **234** by a comparatively fast movement, see the dashed line illustrating the position P of the shifting element **235**.

The dot/dash line shows the position P of the shifting element **235** during a movement in which the shifting element **235** moves into the blocking position **231**. The force F for moving the shifting element **235** is controlled as the target variable Z. In another embodiment, the target variable Z may also comprise a velocity and/or a position P of the shifting element **235**.

Before the shifting element **235** reaches the blocking position **231** in FIG. **2**, the shifting element **235** is moved in a force-controlled manner, as indicated by the curvature of the position P. The force F as target variable Z is the control variable **255** for the movement of the shifting element **235** and is shown as a thick (bold) solid line. Before the shifting element **235** and the further shifting element **236** mesh, the force F is reduced to enable smooth and thus gentle and quiet meshing and shifting.

However, the shifting element **235** reaches the blocking position **231**. The shifting element **235** remains in the blocking position **231** for a short time (illustrated by the dot/dash line being flat at **237** in FIG. **2**). The position P is detected in the blocking position **231** for a time interval and thus corresponds to the state **237** that the shifting element **235** is in the blocking position **231**. To be able to resolve and vacate the blocking position **231** as quickly as possible, an oscillation O of the force F moving the shifting element **235** is imposed on the movement of the shifting element **235**. The force F with the oscillation O forms the control variable **255** for the movement of the shifting element **235** and is shown as a thin (fine) solid line. The oscillation O is a periodic change of the force F about the target variable Z. The force F thus changes in an alternating manner and is alternatingly smaller and larger than the target variable Z.

The oscillation O has an amplitude and a frequency H. The oscillation O and, in particular its frequency H, depends on the speed difference DH between the shifting element **235** and further shifting element **236**. In the force-controlled case according to FIG. **2**, the control signal **255** is determined such that the maximum force F, that is, the sum of the amplitudes of the target variable Z and the oscillation O, falls below a force threshold FT. Alternatively, the position P of the shifting element **235** can also be controlled via a velocity and/or position control. Then the amplitude and/or frequency are determined such that a force resulting from the oscillation remains below the force threshold FT.

After the shifting element **235** has vacated the blocking position **231**, the shifting element **235** is moved in a force-controlled manner, as indicated by the curvature of the position P (increasing from the flat portion of the dot/dash line indicated at **237**) and the solid line as an adjusted control variable **255'**. The vacating of the blocking position **231** can be determined by way of the position P and/or the velocity of the shifting element **235**. The shifting element **235** now moves into the end position **233**. The force F as target variable Z no longer has any oscillation O. The oscillation O is stopped when the blocking position **231** is vacated.

FIG. **3** shows a schematic illustration of a sequence of a method **100** according to one embodiment of the present disclosure. The method **100** is a method **100** for shifting an electrically shiftable transmission **230** for a vehicle **200***a*, in particular a commercial vehicle **200***b*. Such a vehicle **200***a*, **200***b* is described with reference to FIG. **1**. A movement profile **300** for shifting an electrically shiftable transmission

**230** is described with reference to FIG. **2**. FIG. **3** is described with reference to FIGS. **1** and **2**.

The method **100** according to FIG. **3** comprises: detecting **110** a state **237** of the shifting element **235** relating to a blocking position **231** of a shifting element **235** of the transmission **230**. The state **237** represents the blocking position **231** and/or the state **237** indicates the blocking position **231** being present resulting from a movement of the shifting element **235** toward the further shifting element **236**.

A control signal **255** is determined **120** for controlling a movement of the shifting element **235** taking into account the detection of the state **237** and/or the blocking position **231**, wherein the control signal **255** defines a target variable Z of the movement of the shifting element **235** and an oscillation O about the target variable Z. The determination **120** of the control signal **255** is carried out such that a force F on the shifting element **235** falls below a force threshold FT. The target variable Z is a force F, a velocity, and/or a position P. The oscillation O defines a frequency H of the movement of the shifting element **235**. The oscillation O is dependent on a speed difference DH between the shifting element **235** and the further shifting element **236** that can be brought into operative connection with the shifting element **235**.

The control signal **255** is output **130** to move the shifting element **235** based on the control signal **255**.

The method **100** further comprises: detecting **140** or determining the vacating of the blocking position **231**.

An adjusted control signal **255'** is determined **150** for controlling the movement of the shifting element **235** after the blocking position **231** has been vacated, wherein the adjusted control signal **255'** defines a target variable Z of the movement of the shifting element **235** without the oscillation O.

The adjusted control signal **255'** is output **160** to move the shifting element **235** based on the adjusted control signal **255'**.

| Reference Symbols (part of the description) | |
|---|---|
| 100 | Method |
| 110 | Detecting |
| 120 | Determining |
| 130 | Outputting |
| 140 | Detecting |
| 150 | Determining |
| 160 | Outputting |
| 200a | Vehicle |
| 200b | Commercial vehicle |
| 205 | Axle arrangement |
| 230 | Electrically shiftable transmission |
| 231 | Blocking position |
| 232 | Shift actuator |
| 233 | Neutral position |
| 234 | End position |
| 235 | Shifting element |
| 236 | Further shifting element |
| 237 | State |
| 238 | Position sensor |
| 250 | Control device |
| 255 | Control signal |
| 255' | Adjusted control signal |
| 300 | Movement profile |
| DH | Speed difference |
| H | Frequency |
| F | Force |
| FT | Force threshold |
| O | Oscillation |

-continued

| Reference Symbols (part of the description) | |
|---|---|
| P | Position |
| t | Time |
| Z | Target variable |

The invention claimed is:

**1**. A method (**100**) for shifting an electrically shiftable transmission (**230**) for a vehicle (**200***a*, **200***b*), the method (**100**) comprising:

detecting (**110**) a state (**237**) of a shifting element (**235**) of the transmission (**230**) relating to a blocking position (**231**) of the shifting element (**235**) relative to a further shifting element (**236**);

determining (**120**) a control signal (**255**) for controlling a movement of the shifting element (**235**) based on the detected state (**237**) and/or the blocking position (**231**), wherein the control signal (**255**) defines a target variable (Z) of the movement of the shifting element (**235**) and an oscillation (O) about the target variable (Z); and

outputting (**130**) the control signal (**255**) and moving the shifting element (**235**) based on the control signal (**255**).

**2**. The method (**100**) according to claim **1**, wherein the state (**237**) represents an actual detected blocking position (**231**) and/or the state (**237**) indicates a predicted blocking position (**231**) resulting from a movement of the shifting element (**235**) toward the further shifting element (**236**).

**3**. The method (**100**) according to claim **1**, wherein the target variable (Z) is a force (F), a velocity, and/or a position (P).

**4**. The method (**100**) according to claim **1**, wherein the oscillation (O) defines a frequency (H) of the movement of the shifting element (**235**).

**5**. The method (**100**) according to claim **1**, wherein the oscillation (O) is dependent on a speed difference (DH) between the shifting element (**235**) and the further shifting element (**236**) configured to be brought into operative connection with the shifting element (**235**).

**6**. The method (**100**) according to claim **1**, wherein the determining (**120**) of the control signal (**255**) is carried out such that a force (F) on the shifting element (**235**) falls below a force threshold (FT).

**7**. The method (**100**) according to claim **1**, wherein the method (**100**) further comprises:

detecting (**140**) a vacating of the blocking position (**231**);

determining (**150**) an adjusted control signal (**255'**) for controlling the movement of the shifting element (**235**) after the vacating of the blocking position (**231**) is detected, wherein the adjusted control signal (**255'**) defines a target variable (Z) of the movement of the shifting element (**235**) without the oscillation (O); and

outputting (**160**) the adjusted control signal (**255'**) and moving the shifting element (**235**) based on the adjusted control signal (**255'**).

**8**. The method (**100**) according to claim **7**, further comprising moving the shifting element (**235**) from the blocking position into a final position (**234**) engaged with the further shifting element (**236**).

**9**. The method according to claim **8**, further comprising, prior to detecting the state (**237**) relating to the blocking position (**231**), moving the shifting element (**235**) from a neutral position (**233**) toward the further shifting element (**236**).

**10**. The method according to claim **9**, further comprising, prior to reaching the blocking position (**231**) with the shifting element (**235**), reducing a value of the target variable (Z) of the control signal (**255**) Z.

**11**. The method according to claim **10**, wherein a value of the target variable of the adjusted control signal (**255'**) after vacating the blocking position (**231**) is less than the value of the target variable of the control signal (**255**) that is achieved prior to reaching the blocking position (**231**).

**12**. A non-transitory computer-readable medium having instructions stored thereon and comprising commands which, when the commands are executed by a computer having a processor, cause the computer to carry out the method (**100**) according to claim **1**.

**13**. A control device (**250**) for a vehicle (**200***a*, **200***b*) with an electrically shiftable transmission (**230**), wherein the control device (**250**) is configured to carry out the method (**100**) according to claim **1**.

**14**. An electrically shiftable transmission (**230**) having a shifting element (**235**) and a control device (**250**) according to claim **13**.

**15**. An axle arrangement (**205**) for a vehicle (**200***a*, **200***b*), wherein the axle arrangement (**205**) includes an electrically shiftable transmission (**230**) according to claim **14**.

**16**. A vehicle (**200***a*, **200***b*) having an axle arrangement (**205**) according to claim **15**.

**17**. A vehicle (**200***a*, **200***b*) having an electrically shiftable transmission (**230**) according to claim **14**.

**18**. The method according to claim **1**, wherein the target variable of the control signal is a force, where the force is held at a constant value during movement of the shifting element (**235**) from a neutral position (**233**) toward the further shifting element (**236**), wherein a position of the shifting element increases at an exponential rate while the force is held constant, wherein the force is reduced from its constant value as the shifting element (**235**) approaches the further shifting element (**236**) and before an initial engagement therebetween.

**19**. The method according to claim **1**, wherein the state (**237**) of the shifting element (**235**) is detected in response to detecting a position of the shifting (**235**) element via a position sensor (**238**), wherein the position of the shifting element (**235**) does not change over a predetermined time interval.

**20**. The method according to claim **1**, wherein the state (**237**) of the shifting element (**235**) indicates the blocking position resulting from a movement of the shifting element (**235**), wherein the blocking position (**231**) is predicted based on a predicted movement of the shifting element (**235**), wherein the movement of the shifting element (**235**) is detected and, based on a prediction of a position of the shifting element, it is determined that a further movement of the shifting element (**235**) will potentially result in the blocking position (**231**).

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